

BookMyShow — Component Descriptions

High-Level System Architecture Component Breakdown

1. API Gateway

Serves as the single secure entry point for all mobile and web traffic. It handles SSL termination, rate limiting, and verifies OAuth tokens before routing requests to downstream microservices.

2. Load Balancer

Distributes incoming HTTP/HTTPS traffic across multiple application server instances using algorithms like Least Connections to prevent bottlenecks and ensure high availability.

3. Application Servers (Core Services)

- **SearchService:** Handles movie catalog browsing, theater listings, and showtime queries by reading from distributed caches and the search engine.
- **BookingService:** Manages the seat reservation lifecycle, state transitions (Available, Held, Booked), and coordinates order creation using distributed locks to prevent double-booking.
- **PaymentService:** Orchestrates payment checkout workflows, interfaces with external payment gateways, processes webhooks, and manages refunds.
- **NotificationService:** Asynchronously consumes order and payment events from the message queue to dispatch booking confirmations and tickets via email and SMS.

4. Databases

- **Relational Database (PostgreSQL):** Stores transactional records requiring strict ACID guarantees, including shows, seat allocations, bookings, and payment records.
- **NoSQL Database (MongoDB):** Stores high-volume and flexible documents such as user preferences, search logs, and activity audit trails.

5. Cache Layer (Redis)

Caches frequently accessed read data such as active movie catalogs and theater show schedules. It also manages temporary seat hold tokens with expiration TTLs to offload traffic from the primary database.

6. Message Queue (Kafka)

Decouples high-throughput booking transactions from downstream processing. It reliably streams events such as BookingCreated and PaymentFailed to notification and analytics consumers.

7. Search Service (Elasticsearch)

Indexes movie catalogs, cinema locations, and showtimes to provide fast, geo-aware text search and filtered queries for user discovery.

8. External Integrations

- **Payment Gateway (e.g., Razorpay / Stripe):** Third-party provider that securely processes credit cards, debit cards, and UPI transactions.
- **Auth Provider:** Third-party identity provider (e.g., Google OAuth / Firebase) that issues and validates JWT tokens for user authentication and authorization.